

Data Flow Diagram (DFD)

Human Development Index (HDI) Prediction Using Machine Learning

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1. Introduction

A Data Flow Diagram (DFD) is a graphical representation that illustrates how data moves through a system. It identifies the sources of data, processing steps, data storage, and the outputs generated by the system.

For the Human Development Index (HDI) Prediction project, the DFD explains how user input is processed by the machine learning model to generate the predicted HDI value and corresponding development category.

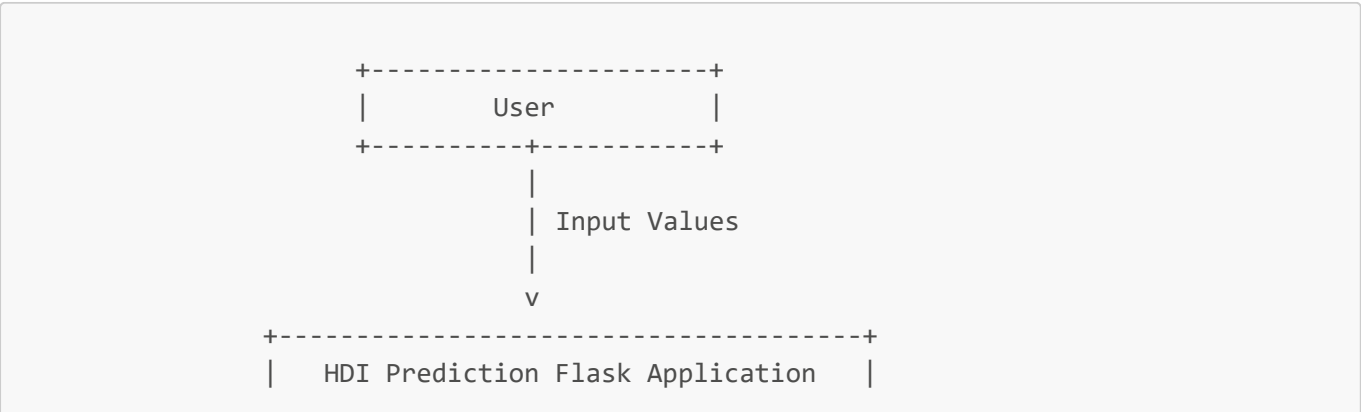
The diagram helps stakeholders understand the overall workflow of the application without focusing on programming details.

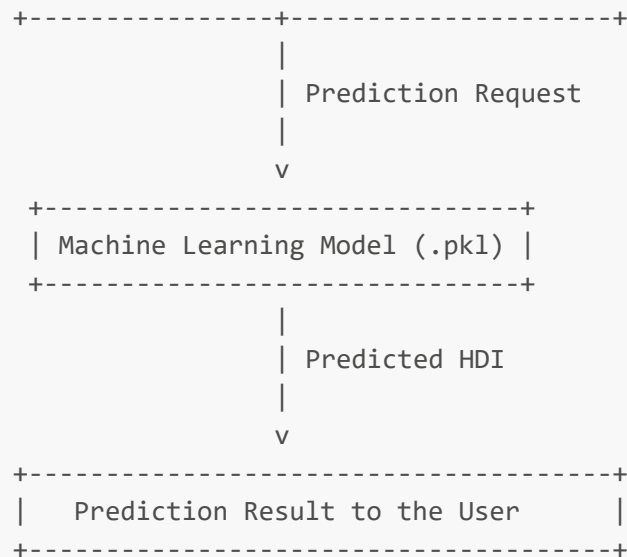
2. Purpose of the Data Flow Diagram

- The Data Flow Diagram was prepared to:
- Understand the flow of information within the system.
 - Show how user inputs are processed.
 - Identify the interaction between the user, Flask application, machine learning model, and dataset.
 - Explain the prediction workflow in a simple and structured manner.
 - Support system documentation for academic and project evaluation.

3. Level-0 Data Flow Diagram (Context Diagram)

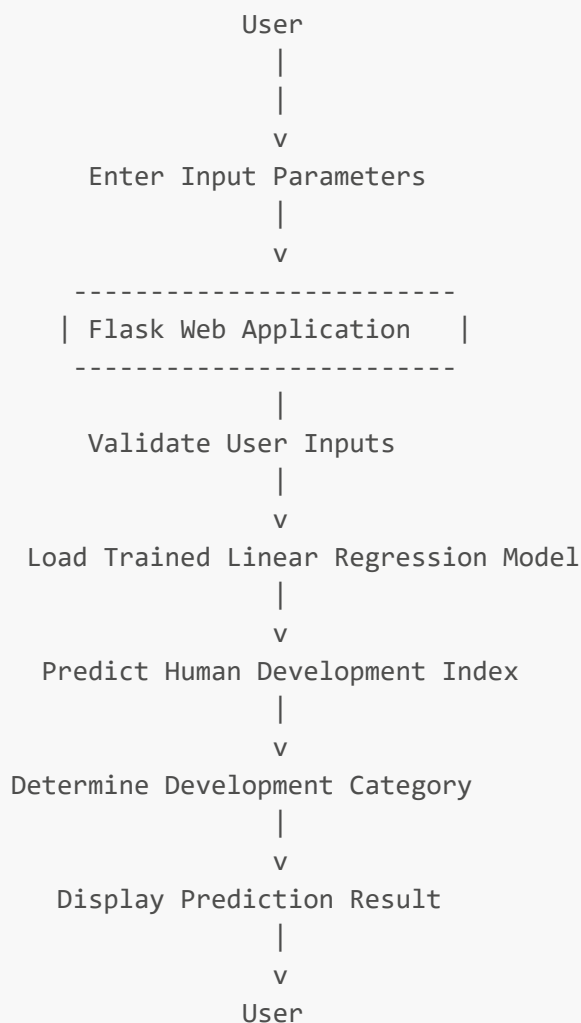
The Level-0 DFD represents the entire system as a single process.





4. Level-1 Data Flow Diagram

The Level-1 DFD provides a more detailed representation of the internal processes.



5. Data Inputs

The prediction system accepts the following input parameters:

Input Feature	Description
Country	Name of the country
Life Expectancy	Average life expectancy at birth
Mean Years of Schooling	Average years of education completed
Expected Years of Schooling	Expected years of education
GNI per Capita	Gross National Income per person

These values are entered through the Flask web interface.

6. Processing Steps

The system performs the following operations:

Step 1

The user enters the required socio-economic information.

Step 2

The Flask application validates the entered values.

Step 3

The trained Linear Regression model is loaded using Joblib.

Step 4

The machine learning model predicts the HDI value.

Step 5

The predicted value is compared with predefined HDI ranges.

Step 6

The development category is determined.

Step 7

The prediction result is displayed on the web page.

7. Data Storage

The system uses the following storage components.

Storage	Purpose
HDI Dataset	Model training
Trained Model (.pkl)	Prediction
Flask Application	User interaction
Static Files	CSS styling
HTML Templates	User Interface

8. Outputs

The system generates the following outputs.

- Predicted HDI Value
 - Human Development Category
 - Validation Messages
 - User-Friendly Prediction Interface
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9. Advantages of the Data Flow Diagram

The DFD provides several benefits.

- Clearly illustrates the movement of data.
 - Simplifies understanding of the application workflow.
 - Helps during system design and development.
 - Useful for debugging and maintenance.
 - Supports project documentation.
 - Makes the prediction process easier to understand.
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10. Conclusion

The Data Flow Diagram provides a clear representation of how data moves through the Human Development Index Prediction System. It begins with user input, continues through validation and machine learning prediction, and ends with displaying the predicted HDI value and development category.

The DFD demonstrates that the application follows a simple, structured, and efficient workflow, making it suitable for academic learning, project documentation, and real-world implementation.